

CSE 208

Digital Logic Design Laboratory

Course Plan

Course Code: CSE 208
Course Title: Digital Logic Design Laboratory
Course Type: Laboratory
Number of Classes: 12
Laboratory Focus: Digital Circuit Design and Implementation

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Academic Year: 2026

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1 Course Overview

CSE 208: Digital Logic Design Laboratory provides hands-on experience with fundamental digital logic circuits. The laboratory is designed to complement theoretical knowledge by allowing students to design, construct, test, and troubleshoot digital circuits using standard logic ICs and laboratory equipment.

The course follows a progressive structure, beginning with basic logic gates and gradually introducing combinational and sequential circuits. Students ultimately integrate their knowledge in a comprehensive practical exercise.

1.1 Course Progression

The laboratory follows the sequence:

**Logic Gates → Universal Gates →
Adders/Subtractors → MUX/DEMUX →
Encoder/Decoder → Flip-Flops → Integrated
Practical**

2 Course Objectives

The main objectives of this laboratory course are to:

1. Develop practical understanding of fundamental digital logic gates.
2. Implement Boolean expressions using standard logic gates and universal gates.
3. Design and verify arithmetic circuits such as adders and subtractors.
4. Understand and implement multiplexers and demultiplexers.
5. Understand the operation and applications of encoders and decoders.
6. Introduce students to sequential logic and flip-flops.
7. Develop practical skills in breadboard-based digital circuit construction.
8. Develop circuit debugging and troubleshooting skills.
9. Enable students to translate a digital system specification into a practical circuit.
10. Integrate multiple concepts to solve comprehensive digital circuit design problems.

3 Course Learning Outcomes (CLOs)

After successful completion of the course, students will be able to:

CLO	Learning Outcome
CLO1	Identify and implement fundamental digital logic gates and verify their truth tables experimentally.
CLO2	Construct Boolean functions using basic and universal logic gates.
CLO3	Design, construct, and verify half/full adders and half/full subtractors.
CLO4	Design and implement multiplexers and demultiplexers for data selection and distribution.
CLO5	Design and implement encoders and decoders and understand their applications.
CLO6	Construct and analyze SR, JK, D, and T flip-flops.
CLO7	Interpret experimental results and identify errors in digital circuits.
CLO8	Apply knowledge of combinational and sequential circuits to solve integrated digital system problems.

4 Course Structure

Class	Topic	Major Practical Activity
1	Basic Logic Gates	Verification of AND, OR, NOT, NAND, NOR, XOR, and XNOR gates.
2	Universal Gates	Implementation of basic gates and Boolean functions using NAND and NOR.
3	Half Adder and Half Subtractor	Design, construction, and verification of half adder and half subtractor.
4	Full Adder	Design and implementation of a full adder using half adders and logic gates.
5	Full Subtractor	Design and implementation of a full subtractor using half subtractors.
6	Multiplexer	Implementation of 2:1 and 4:1 multiplexers and Boolean function realization.
7	Demultiplexer	Implementation of 1:2 and 1:4 demultiplexers.
8	Encoder	Implementation of 4:2 encoder and introduction to priority encoders.
9	Decoder	Implementation of 2:4 and 3:8 decoders and function realization.
10	SR and JK Flip-Flops	Construction and verification of SR and JK flip-flops.
11	D and T Flip-Flops	Construction and verification of D and T flip-flops.
12	Comprehensive Practical	Integrated digital circuit design, implementation, testing, and troubleshooting.

5 Detailed Laboratory Class Plan

5.1 Class 1: Basic Logic Gates

Topics

- Digital logic and binary signals
- Logic HIGH and LOW
- AND gate
- OR gate
- NOT gate
- NAND gate
- NOR gate
- XOR gate
- XNOR gate
- Truth tables

Practical Activities

1. Identify the required logic ICs.
2. Set up the breadboard.
3. Connect VCC and GND correctly.
4. Construct each basic logic gate.
5. Apply different input combinations.
6. Observe the output using LEDs.
7. Record experimental results.
8. Compare experimental outputs with theoretical truth tables.

Experiment

Verification of the truth tables of basic logic gates.

Expected Outcome

Students will be able to identify and experimentally verify the operation of fundamental logic gates.

5.2 Class 2: Universal Gates

Topics

- Universal gates
- NAND as a universal gate
- NOR as a universal gate
- NAND implementation of NOT
- NAND implementation of AND
- NAND implementation of OR
- NOR implementation of NOT
- NOR implementation of AND
- NOR implementation of OR
- Boolean function realization using universal gates

Practical Activities

Students will implement:

1. NOT using NAND.
2. AND using NAND.
3. OR using NAND.
4. NOT using NOR.
5. AND using NOR.
6. OR using NOR.
7. A given Boolean expression using only NAND gates.
8. A given Boolean expression using only NOR gates.

Experiment

Realization of basic logic gates using universal gates.

5.3 Class 3: Half Adder and Half Subtractor

Topics

- Binary addition
- Half adder
- Sum and Carry
- Half subtractor
- Difference and Borrow
- Truth tables
- Boolean expressions

Practical Activities

1. Derive the truth table of a half adder.
2. Implement the half adder.
3. Verify Sum and Carry outputs.
4. Derive the truth table of a half subtractor.
5. Implement the half subtractor.
6. Verify Difference and Borrow outputs.

Experiment

Design and implementation of half adder and half subtractor.

5.4 Class 4: Full Adder

Topics

- Limitation of half adder
- Carry-in
- Full-adder truth table
- Sum and Carry-out
- Full adder using two half adders
- Cascading full adders

Practical Activities

1. Construct a full adder using two half adders.

2. Apply all combinations of inputs.
3. Verify Sum and Carry-out.
4. Compare the full adder with the half adder.
5. Discuss cascading full adders for multi-bit addition.

Experiment

Design and implementation of a full adder using half adders.

5.5 Class 5: Full Subtractor

Topics

- Limitation of half subtractor
- Borrow-in
- Full-subtractor truth table
- Difference
- Borrow-out
- Full subtractor using two half subtractors

Practical Activities

1. Construct a full subtractor.
2. Apply all combinations of inputs.
3. Verify Difference and Borrow-out.
4. Compare full adder and full subtractor structures.

Experiment

Design and implementation of a full subtractor.

5.6 Class 6: Multiplexer

Topics

- Multiplexer concept
- Data selection
- Select lines
- 2:1 multiplexer
- 4:1 multiplexer

- Truth tables
- Boolean function realization using MUX

Practical Activities

1. Construct a 2:1 multiplexer.
2. Verify its truth table.
3. Construct a 4:1 multiplexer.
4. Verify its operation.
5. Implement a Boolean function using a multiplexer.

Example

Implement:

$$F(A, B, C) = \Sigma m(1, 2, 5, 7)$$

using a suitable multiplexer configuration.

Experiment

Design and implementation of multiplexers.

5.7 Class 7: Demultiplexer

Topics

- Demultiplexer concept
- Data distribution
- Select lines
- 1:2 DEMUX
- 1:4 DEMUX
- Truth tables
- Applications

Practical Activities

1. Construct a 1:2 demultiplexer.
2. Verify its truth table.
3. Construct a 1:4 demultiplexer.
4. Test different select-line combinations.
5. Verify that data reaches only the selected output.
6. Compare MUX and DEMUX operations.

Experiment

Design and implementation of demultiplexers.

5.8 Class 8: Encoder

Topics

- Encoder concept
- Binary encoding
- 4:2 encoder
- Truth table
- Logic implementation
- Multiple active inputs
- Priority encoder

Practical Activities

1. Construct a 4:2 encoder.
2. Apply valid input combinations.
3. Observe binary outputs.
4. Verify the truth table.
5. Discuss the limitation of ordinary encoders.
6. Introduce the concept of a priority encoder.

Experiment

Design and implementation of an encoder.

5.9 Class 9: Decoder

Topics

- Decoder concept
- Binary decoding
- 2:4 decoder
- 3:8 decoder
- Enable input
- Truth table
- Logic implementation
- Boolean function realization

Practical Activities

1. Construct a 2:4 decoder.
2. Verify all output combinations.
3. Implement or demonstrate a 3:8 decoder.
4. Investigate the enable input.
5. Implement a Boolean function using a decoder.

Experiment

Design and implementation of decoders.

5.10 Class 10: SR and JK Flip-Flops**Topics**

- Combinational vs. sequential circuits
- Memory elements
- Clock signal
- SR flip-flop
- Set and Reset operations
- Invalid state
- JK flip-flop
- Race-around condition
- Truth tables

Practical Activities**Part A: SR Flip-Flop**

1. Construct the SR flip-flop.
2. Apply Set and Reset inputs.
3. Verify Hold and invalid states.
4. Record the outputs.

Part B: JK Flip-Flop

1. Construct or connect a JK flip-flop.
2. Verify Set, Reset, Hold, and Toggle operations.
3. Observe the effect of the clock.

Experiment

Design and verification of SR and JK flip-flops.

5.11 Class 11: D and T Flip-Flops**Topics**

- D flip-flop
- Data storage
- T flip-flop
- Toggle operation
- Truth tables
- Clocked operation
- Frequency division
- Applications of flip-flops

Practical Activities**Part A: D Flip-Flop**

Students verify the relationship:

$$Q_{\text{next}} = D$$

Part B: T Flip-Flop

Students verify:

$$T = 0 \Rightarrow Q_{\text{next}} = Q$$

$$T = 1 \Rightarrow Q_{\text{next}} = \overline{Q}$$

Students will also demonstrate the use of a T flip-flop as a frequency divider.

Experiment

Design and verification of D and T flip-flops.

5.12 Class 12: Comprehensive Practical

Objective

The final laboratory session evaluates the student's ability to independently analyze, design, implement, test, and troubleshoot a digital circuit.

Required Activities

Students should:

1. Analyze the given problem.
2. Identify the inputs and outputs.
3. Develop the truth table.
4. Derive the Boolean expression.
5. Simplify the expression where appropriate.
6. Select appropriate ICs/components.
7. Draw the circuit diagram.
8. Construct the circuit.
9. Test the circuit using different input combinations.
10. Record the results.
11. Troubleshoot any circuit errors.
12. Explain the operation during the viva.

Possible Comprehensive Problems

1. Design a 4-bit binary adder using full adders.
2. Implement a Boolean function using a 4:1 multiplexer.
3. Implement a Boolean function using a decoder.
4. Design an encoder-decoder system.
5. Design a simple binary counter using flip-flops.
6. Design a digital control circuit from a given specification.

6 Recommended Laboratory Report Format

Each laboratory report should contain the following sections:

1. Experiment Number
2. Experiment Title
3. Date
4. Objective
5. Required Components
6. Theoretical Background
7. Truth Table
8. Boolean Expressions
9. Circuit Diagram
10. Procedure
11. Experimental Results
12. Discussion
13. Error Analysis
14. Conclusion
15. Viva Questions

7 Required Laboratory Equipment and Components

7.1 Basic Equipment

- Digital logic trainer or breadboard
- Regulated DC power supply
- Digital multimeter
- Oscilloscope (where applicable)
- Function/signal generator (where applicable)
- Connecting wires/jumper wires
- LEDs
- Resistors
- Push buttons/switches

7.2 Suggested Logic ICs

Function	Example IC	Purpose
AND	7408	Four 2-input AND gates
OR	7432	Four 2-input OR gates
NOT	7404	Six NOT gates
NAND	7400	Four 2-input NAND gates
NOR	7402	Four 2-input NOR gates
XOR	7486	Four 2-input XOR gates
MUX	74153 / 74151	Multiplexer implementation
DEMUX	74138 / suitable DEMUX IC	Data distribution / decoding
Encoder	Suitable encoder IC	Encoding experiments
Decoder	74138 / 7447	Decoder implementation
JK FF	7476 / suitable IC	JK flip-flop experiments
D FF	7474	D flip-flop experiments

Note: IC selection may be adjusted according to the available laboratory equipment and the chosen logic family (e.g., TTL or CMOS).

8 Digital Circuit Troubleshooting

Students should be trained to systematically identify circuit errors.

8.1 Common Problems

- Incorrect IC orientation
- Incorrect VCC/GND connections
- Loose jumper wires
- Floating inputs
- Incorrect pin connections
- Incorrect LED polarity
- Missing current-limiting resistors
- Damaged ICs
- Incorrect truth-table interpretation

- Active-low input/output misunderstanding
- Clock signal problems in sequential circuits

8.2 Recommended Debugging Procedure

Students should follow:

1. Turn off the power.
2. Verify the circuit diagram.
3. Verify IC orientation.
4. Verify VCC and GND connections.
5. Check every input and output connection.
6. Check for floating inputs.
7. Check LED and resistor connections.
8. Power the circuit again.
9. Test each section independently.
10. Compare observed results with the theoretical truth table.

9 Assessment Scheme

A recommended assessment distribution is:

Assessment Component	Weight
Class Attendance	10%
Class Performance/Attire	10%
Lab Reports	20%
Field Assessment and Viva	20%
Final Exam	40%
Total	100%

10 Suggested Viva Assessment Areas

The viva should evaluate both theoretical understanding and practical skills.

10.1 Logic Gates

- What is a universal gate?
- Why are NAND and NOR called universal gates?
- What is the difference between XOR and OR?
- What happens when one input of an AND gate is LOW?

10.2 Adders and Subtractors

- Why cannot a half adder handle a carry-in?
- What is the difference between carry and borrow?
- How can a full adder be constructed using two half adders?
- How can multiple full adders be cascaded?

10.3 MUX and DEMUX

- Why is a multiplexer called a data selector?
- How many select lines are required for an 8 : 1 MUX?
- What is the function of a demultiplexer?
- What is the relationship between the number of select lines and outputs?

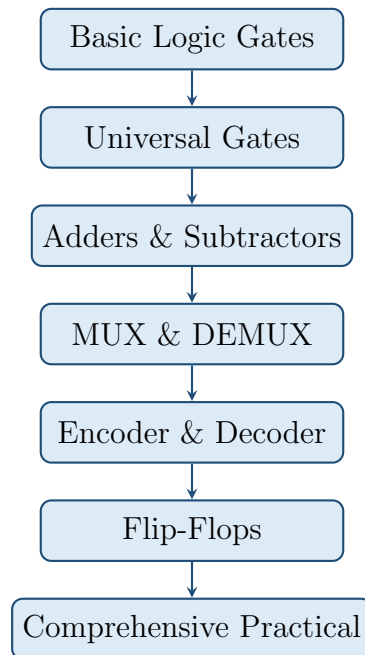
10.4 Encoder and Decoder

- What is an encoder?
- What is a decoder?
- What is a priority encoder?
- What is the purpose of an enable input?

10.5 Flip-Flops

- What is the difference between combinational and sequential circuits?
- What is the purpose of a clock?
- What is the invalid state of an SR flip-flop?
- What is the difference between JK and SR flip-flops?
- How does a D flip-flop store data?
- What happens when $T = 1$ in a T flip-flop?

11 Course Concept Map



Conclusion

CSE 208: Digital Logic Design Laboratory is structured to provide a progressive practical foundation in digital system design. Students begin with individual logic gates, advance through combinational circuits such as adders, multiplexers, encoders, and decoders, and conclude with sequential circuits based on flip-flops and an integrated comprehensive practical.

The emphasis throughout the course is on **design, implementation, verification, troubleshooting, and practical problem solving.**